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Magneto-Rayleigh-Taylor Instability Experiments on a Dense Z-Pinch Plasma

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We report experiments to explore the magneto-Rayleigh-Taylor instability in plasmas generated by subjecting 400 nm-thick Al foils to 0.6-0.7 MA pulsed currents. The current generator is a linear transformer driver (LTD) with a voltage of 70-100 kV, risetime of 100 ns and pulse duration of 200 ns. The planar foil is located between two planar return current plates. Foil plasma acceleration can be adjusted by offsetting the initial foil position from the axis. Instability growth is diagnosed by sub-ns laser shadowgraphy. Results show exponential growth consistent with theory and modelling with fastest growth corresponding to the maximum plasma acceleration. Seeding of a preferred, instability wavelength is accomplished by laser micromachining target foils. Theory and simulation will be presented to elucidate the effects of axial magnetic fields and magnetic shear.

1. Introduction and Experimental Configuration

The magneto-Rayleigh-Taylor instability (MRT) [1,2] is ubiquitous in nature, e.g., Crab nebula.[3] MRT is also a limiting factor in magnetized fusion concepts, particularly Sandia National Laboratory's MagLIF concept in which an imploding cylindrical liner heats and compresses a hot, magnetized target plasma.[4] The driving mechanism for MRT is the magnetic field (light fluid) that accelerates the plasma (heavy fluid), analogous to water above oil in a gravitational field.

In order to excite MRT in a dense plasma an experiment was designed in a planar plasma geometry utilizing a 400 nm-thick, 1-cm-wide, 2-cm long foil enclosed on two sides by planar return-current electrodes.[5] The system is depicted schematically in Fig. 1. If the foil is centered between the electrodes there exists no net force on the ablated foil plasma. However, when the foil is offset, the foil plasma is accelerated towards the center.

A 1-MA, 100 kV Linear Transformer Driver (LTD) generates the foil ablation and acceleration. This LTD consists of parallel connections of 40 "bricks" each consisting of two 40 nH capacitors and one spark-gap switch. A magnetically-insulated transmission line transmits the electrical pulse to the load region. A sub-ns Nd-

YAG or Ti-sapphire laser captures a shadowgraphic image of the plasma viewed parallel to the foil, either perpendicular or parallel to the current flow. Fourier analysis of the shadowgrams yields the instability spectrum and a temporal sequence of shadowgrams on separate shots. The instability spectrum is typically noisy making it difficult to characterize the growth rate. In order to seed the instability wavelength, arrays of laser ablated (30 micron) holes are micromachined in macroscopic strips of typically 1-mm or 2-mm period, as shown in Figure 2c. The varying ablation rate of these strips "seeds" the MRT instability wavelength.

2. MRT Experiments

2.1. Experiments Performed

A series of experiments have been performed to measure the growth rate of MRT as a function of plasma acceleration. Two experimental cases were run. In the first, the foil initial position was offset by 1.1-mm (large-offset) in order to achieve the desired acceleration to drive the instability. In the second, baseline-case, the foil was centered between the electrodes within 0.13-mm (small-offset).

In a later series of experiments, the foils were micromachined, prior to shots, with arrays of holes with macroscopic strips spaced by 1-mm or 2-mm. These strips were intended to seed the MRT.

2.2. Experimental Results

Early time shadowgraphy data show the growth of the electro-thermal (ET) instability, the spectrum of which peaks at short wavelengths. Later exponential temporal growth of longer wavelength perturbations from the MRT was observed experimentally. In qualitative agreement with theory, the fastest instability growth was observed in the large-offset foil case, which exhibited the highest acceleration. The decelerating side of the foil exhibited slower growth. Both sides of the foil plasma showed intermediate growth rate for the small-offset foil (centered) case. For all unseeded cases the instability wavelength was noisy and temporally evolved from short wavelength (ET) to long-wavelength (MRT).

Seeded foil MRT experiments showed preferred wavelengths of perturbation growth close the foil-seeding wavelength, as depicted in Figure 2. Future experiments will utilize these seeded foils to accurately measure the growth rate/ e-folding times of instabilities versus acceleration at a single wavelength.

2.3. Theory of MRT Stabilization by Axial Magnetic Field and Magnetic Shear

A pre-existing axial magnetic field, together with the radial dependence of the azimuthal magnetic field within the liner, constitutes a strong magnetic shear, offering the interesting possibility of shear stabilization of MRT.[6] While MRT may not be completely eliminated with the strong magnetic shear, magnetic shear does reduce the MRT growth rate in general. The feedthrough factor [2] is found to be insensitive to the magnetic shear.

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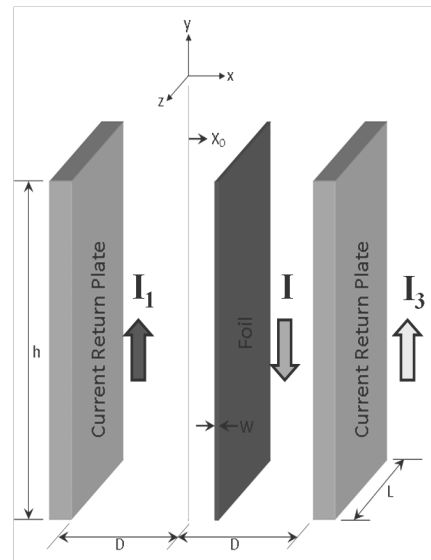


Figure 1. Schematic illustration of experiment for ablation/ acceleration of foil ablation plasma.

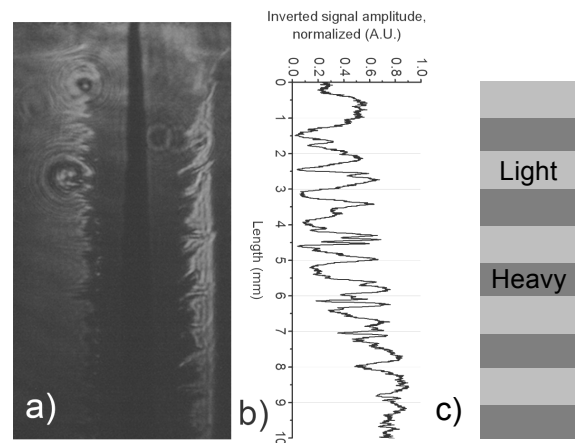


Figure 2. (a) Laser shadowgraph of 1-mm-period micromachined foil during shot 443. (b) Spatially integrated line-out of the right side of the foil. (c) Micromachining schematic of the foil, showing variation in mass density. Peak current was 600 kA in 160 ns, and the image was taken at 156 ns. The effect of seeding is seen with a dominant single wavelength is observable around 0.75 mm.